

8. The table shows some physical properties of Group 6 elements.

Element	Melting point (°C)	Boiling point (°C)	Density (g/cm ³)	Electrical conductor
oxygen	−219	−183	0.0014	no
sulfur	115	445	2.0	no
selenium	221	685	4.8	semi-conductor
tellurium	450	988	6.2	semi-conductor

(a) (i) Describe the trend in the melting points of the Group 6 elements. [1]

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(ii) Give the physical state of selenium at 400 °C. Give a reason for your choice. [2]

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(iii) Explain why it is difficult to classify selenium as either a metal or a non-metal. [1]

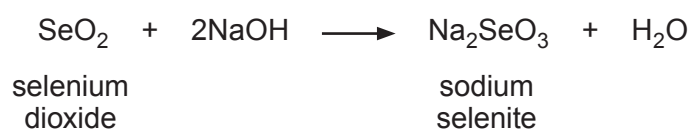
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(b) Selenium dioxide reacts with sodium hydroxide to produce sodium selenite and water.



(i) Calculate the relative formula mass (M_r) of sodium selenite, Na_2SeO_3 . [1]

$$A_r(\text{Na}) = 23 \qquad A_r(\text{Se}) = 79 \qquad A_r(\text{O}) = 16$$

$M_r = \dots\dots\dots$

(ii) Calculate the percentage by mass of selenium in sodium selenite. [2]

Percentage = $\dots\dots\dots$ %

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END OF PAPER



FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^-
ammonium	NH_4^+	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^-
calcium	Ca^{2+}	fluoride	F^-
copper(II)	Cu^{2+}	hydroxide	OH^-
hydrogen	H^+	iodide	I^-
iron(II)	Fe^{2+}	nitrate	NO_3^-
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^+	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^+		
silver	Ag^+		
sodium	Na^+		
zinc	Zn^{2+}		



THE PERIODIC TABLE

Group 1 2 3 4 5 6 7 0



24

1 H Hydrogen 1

7 Li Lithium 3	9 Be Beryllium 4
23 Na Sodium 11	24 Mg Magnesium 12
39 K Potassium 19	40 Ca Calcium 20
86 Rb Rubidium 37	88 Sr Strontium 38
133 Cs Caesium 55	137 Ba Barium 56
223 Fr Francium 87	226 Ra Radium 88
	227 Ac Actinium 89

11 B Boron 5	12 C Carbon 6	14 N Nitrogen 7	16 O Oxygen 8	19 F Fluorine 9	20 Ne Neon 10
27 Al Aluminium 13	28 Si Silicon 14	31 P Phosphorus 15	32 S Sulfur 16	35.5 Cl Chlorine 17	40 Ar Argon 18
70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36
115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54
204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	210 Po Polonium 84	210 At Astatine 85	222 Rn Radon 86

59 Co Cobalt 27	56 Fe Iron 26	59 Ni Nickel 28	63.5 Cu Copper 29	65 Zn Zinc 30
103 Rh Rhodium 45	101 Ru Ruthenium 44	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48
186 Re Rhenium 75	184 W Tungsten 74	192 Ir Iridium 77	197 Au Gold 79	201 Hg Mercury 80

Key

Ar	relative atomic mass
Symbol	
Name	
Z	atomic number